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On the Graph-Based Extension of the Set Based Model: A New Approach on Graph Representation and Model Ensemble.

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Abstract

The purpose of this paper is to present a set of algorithmic improvements upon an extension of the Set-Based Model ([Kalogeropoulos et al \(2020\)](#)) that focuses on the dependence among document terms employing graph representations. A graph-based approach to document representation can positively affect the information retrieval process due to the ability of graphs to capture the syntactic notion and, in some cases, the semantic relationship among document terms. The aforementioned model generates complete graphs; thus, each document term will be interdependent with the rest. Consequently, an interdependent segment or multiple parts of a document called a window is defined, in which the graph creation algorithms are applied. The proposed methods aim to approximate the window size by exploiting the document length. Moreover, an attempt to create multiple windows is made, considering the relationship between a sentence and a paragraph, which is reflected in the semantic importance of nodes and edges. An attempt to tackle the stop-word detection problem on bridge nodes is made by implementing algorithmic schemes that use core decomposition ([Seidman \(1983a\)](#)) to identify the importance of such nodes in a sample of the corpus collection. Finally, a simple reranking scheme and an ensemble voting technique are implemented to enhance model performance on queries where the proposed approach lacks performance. The experimental analysis made on multiple document collections exhibits performance improvements and in some queries, our

approach outperforms even state-of-the-art models such as BM25 (Robertson et al (2004)) and ColBERT (Khattab and Zaharia (2020)).

Keywords: Information Retrieval Models, Set-Based Model, Graphical Document Representation, Stopword Detection, Model Ensemble, Borda Count

1 Introduction

The main focus of information retrieval is the creation of effective and efficient models that comply with the user’s information-seeking needs, which are usually expressed in the form of unstructured queries. In this study, following the approach of Kalogeropoulos et al (2020), an effort is made to alleviate the connection among terms in the textual graphs that arise in their proposed extension of the Set-Based model (Possas et al (2002), Pôssas et al (2005)) by implementing appropriate text partitions termed *windows*, and combining appropriately the resulted scheme with state-of-the-art approaches.

The gist of Kalogeropoulos et al (2020) scheme was based on the hypothesis that each term on a document has an equal bidirectional relationship with each other, thus forming a complete graph for each document. Terms are represented as nodes on the graph and their connections as edges. Linguistically, though, one can easily understand that the starting hypothesis is simplistic, thus it is imperative to propose methods to restrain the intradocument relations.

Graph theoretic models on information retrieval emerged around 1957 (Firth (1957)). Since then, several graphical formalisms have been implemented to express textual data as such graphs. According to, Medeiros Soares et al (2005) the data is separated into syllables of words, creating a graph, in which, each node depicts a syllable and an edge is created between nodes (syllables) that exist in the document. Bordag et al (2003) noticed the semantic relationship of terms in a well-formed text segment, thus suggesting a similarity measure for collocations inside sentences. The proposed graph is bidirectional. Each term is a node on the graph and edges stand for the collocation probability of those terms, which is expressed by the thickness of the edge. Furthermore, Ferrer i Cancho et al (2007) and Ferrer i Cancho et al (2004) focused on terms of syntactic relationships, when creating the graph. Finally, given a document collection, Blanco and Lioma (2012) created two different aspects of co-occurrence text graphs. The first one is undirected and an edge between two nodes—terms is formed if they exist within a window of terms. The term weights are computed similarly to TextRank (Mihalcea and Tarau (2004)), which is a variation of PageRank (Page et al (1999)) or with a degree-based metric, called TextLink. Secondly, they implemented a directed co-occurrence graph with grammatical constraints; these constraints are expressed by employing part-of-speech (PoS) tagging. In such a graphical approach, the direction of every edge defines a grammatical dependency between the respective terms.

Rousseau and Vazirgiannis (2013) proposed Graph of Words, a textual representation and a degree-based weighting scheme, which implements a w -sized sliding window,

where w is the number of terms that the window consists of. The graph is undirected, even though, a directed case was implemented and found useful in applications such as document summarization or phrasal indexing. Terms are represented as nodes. An edge is formed between two nodes if they exist in the same window. The edge weight reflects the number of nodes co-existing in a window. Finally, for each term in a document, a weight is calculated, which depends on normalized node weight in the respective graph, as well as, the term’s inverse document frequency in the collection, having many similarities with the BM25 model (Robertson and Sparck Jones (1977), Robertson and Walker (1994), Robertson et al (2004), Sparck Jones et al (2000)).

In further research, Rousseau and Vazirgiannis (2015) approached the problem of keyword detection using core decomposition (Seidman (1983a)). By retaining nodes on the main core of each textual graph, they detected keywords of the respective document. In a later step, they introduced methods of estimating important nodes using subgraphs other than the main core. As a first attempt, they focus on dense cores or trusses (Cohen (2004)), ultimately proposing a node ranking that depends on the sum of core numbers in which each neighbor belongs for a given node (Tixier et al (2016)).

One important aspect and method of performance enhancement in not only information retrieval models but also clustering or classification is *the model ensemble*. Such approaches focus on the combination/fusion of multiple base models resulting in an outcome. Some of the oldest, yet not deprecated methods are associated with election processes. Voting methods, such as Borda (de Borda (1781)) or Condorcet (Young (1988), Tyomkin and Kurland (2023)) counts are implemented to merge lists of rankings derived from information retrieval models, considering such models as voters and documents as candidates. Fusion models can capture the diversity of base estimators’ characteristics, exhibit robustness on models’ failures, and are capable of amplifying retrieval precision without many compromises on performance. In this paper, we will employ the Borda count voting method, although in the bibliography, the Condorcet is implemented as well (Markovits et al (2012), Anava et al (2016), Rabinovich et al (2014)).

Similarly, a performance enhancement can be achieved by reranking, an approach that is mainly used in combination with neural retrieval models (Lin et al (2022)). The main idea behind the reranking process is to use a two-phase ranking scheme that employs two different ranking algorithms in series. The first ranking algorithm produces a ranking list, which will be processed by the second scheme. Note that we can repeat this process by employing more reranking phases.

In most of the above-related work, text segmentation using windows is one of the most prominent solutions. Here, we will try to embed that notion upon the Graph-Based extension of the Set-Based model. We will try to estimate the effect of window sizes, following the logical linguistic partition of sentence, paragraph, and a combination of the above, implementing multiple windows.

The remainder of this paper is organized as follows. In Section 2, the graph-based extension of the Set-based model is summarized, to acknowledge the focal points and novelties, which will be discussed in Section 3. In that part, an alternative graph creation will be presented, as well, as multiple windows implementations. Furthermore,

in that section we will elaborate on key nodes, their importance, the implemented reranking technique and the implemented ensemble method, which will be tested in Section 4, considering multiple collections having a variety of document and query structures. Finally, in the last section, our conclusion will be drawn and we will present future contribution aspects and directions.

2 Preliminaries and methods

As mentioned before, [Kalogeropoulos et al \(2020\)](#) proposed an information retrieval model employing graphs. Each document is mapped to a complete graph, called *Rational Path Graph*. Each term is related to the rest as a consequence of the graph’s completeness. Every term of a given document is represented as a node in the graph and a relationship between two nodes as a weighted edge (out-edge) having weight W_{out} . Additionally, each node is characterized by a self-loop weighted edge (in-edge) with weight W_{in} . The weights of each out-edge and in-edge are calculated concerning the term frequency of the respective nodes, taking into consideration the lemmas that follow.

2.1 Rational Path Graph

In their work [Kalogeropoulos et al \(2020\)](#), proposed a graph creation algorithm. As an input, they used an inverted list, which contains terms and their respective term frequencies in the documents thus forming a Rational Path Graph as an output. In our work, we will propose an alternative method of constructing the same graph, with different representations as an input.

Definition 1.

$$W_{in} = \frac{tf_i \cdot (tf_i + 1)}{2} \quad (1)$$

Definition 2.

$$W_{out} = tf_i \cdot tf_j, \text{ WHEN } N_i \neq N_j \quad (2)$$

2.2 Union Graph

After a document is processed and the rational path graph is completed, a collection-sized graph is created as the union of each separate document’s graph ([Sonawane and Kulkarni \(2014\)](#)). The graph union is merging two or more graphs without any information loss to the initial corpus. If a document collection D with documents $\{d_1, d_2, \dots, d_n\}$ is considered and their respective graph representatives are $\{G_1, G_2, \dots, G_n\}$ then the graph union of the collection D is represented as

$$G_D = \bigcup_{i=1}^n G_i \quad (3)$$

In practice, as shown in figure 1 when merging separate graphs into one union graph G_D new nodes or edges are included in the new graph, and on co-existing edges. The

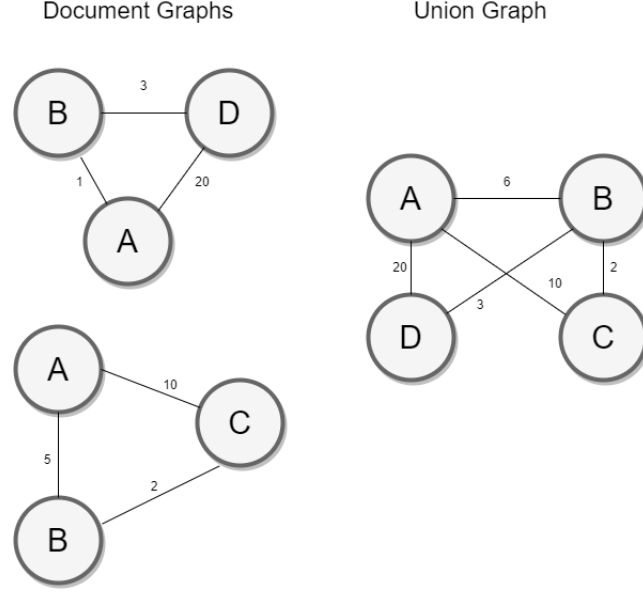


Fig. 1 Union Graph Example

weight of an edge on the union graph is calculated as the sum of individual weights. For example, in figure 1 the edge (A, B) exists in both document graphs, thus the weight of that edge is calculated as 6 in their union. Similarly, nodes C and D are included along with their associated edges.

At this point, following the work of [Rousseau and Vazirgiannis \(2015\)](#) and [Tixier et al \(2016\)](#) on keyword estimation, an attempt to augment the weight of such words is made, implementing in a similar manner core decomposition on document graphs. In the context of graph theory, core decomposition of a graph involves partitioning the nodes into layers known as cores, where each layer represents a subset of nodes with a certain level of connectivity gauged by each node degree on that sub-graph ([Seidman \(1983b\)](#)). Due to the graph completeness, though, a pruning process is imperative, because on a complete graph, all nodes belong to the main core, by definition.

At this point, an average edge weight is calculated on the rational path graph, and edges that are less than a threshold are removed. That limit is calculated as $p \cdot \text{Average edge weight}$, where p is a heuristic percentage variable. Finally, every edge related to a keyword node is multiplied by an importance variable h .

After completion of processing the document collection, a derived of the union graph weight is calculated for each node k , as shown in the following equation.

$$nw_k = \log \left(1 + a \frac{W_{out_k}}{(W_{in_k} + 1)(Cn_k + 1)} \right) \log \left(1 + b \frac{1}{Cn_k + 1} \right) \quad (4)$$

where, W_{out_k} is the sum of the weights of all out-edges of node k , W_{in_k} is the weight of node's k in-edge, Cn_k stands for the number of its neighbors, and a, b are

the variables which affect the importance of graph-based extension against the Set-Based model original weights. Thereafter, a simple inverted index can be created, which contains the calculated weights, as well as, supplementary information used by the Set-Based model's weighting process.

2.3 Graphs and Set-Based model

The retrieval process is handled by the Set-Based model. As mentioned, the notion of termsets, related to query terms, contributes highly to complexity reduction, due to processing significantly lower data volume (Possas et al (2002), Pôssas et al (2005)). It is important to notice that the model implements association rule mining algorithms (Agrawal and Srikant (1994)), to combine two frequent termsets in a different element of each set, thus creating a new one, which if frequent, will be considered a termset of the model. The frequency threshold is user-defined and contributes to decreasing the complexity even more.

The set-based model uses a termset frequency (TS) and inverse document frequency (IDF) based metric for termset weights. The graph-based extension introduces a new measure, (tnw), in order to combine the graph-derived weights. Therefore, for a termset S_i consisting of k terms the weight is calculated as the product of those term weights as determined on the union graph.

$$tnw_{S_i} = \prod_{k \in S_i} nw_k \quad (5)$$

The new measure acts multiplicatively to the set-based model's weighting process, as shown in the following equation. In this equation, Sf_{ij} represents the term frequency of the set S_i in document j and $\frac{N}{dS_i}$ denotes the inverse document frequency of termset S_i , which will be used as the query scoring function.

$$W_{S_{ij}} = (1 + \log Sf_{ij}) \cdot \log \left(1 + \frac{N}{dS_i} \right) \cdot tnw_{S_i} \quad (6)$$

$$W_{S_{iq}} = \log \left(1 + \frac{N}{dS_i} \right) \quad (7)$$

Finally, the document ($\vec{d}_j$) and query ($\vec{Q}$) vectors are formed having as the size of at most 2^n elements, where n is the number of unique terms that the query consists of. It is imperative to notice that a collection-wise termset calculation would be non-realistic since it is computationally expensive because the lexicon size N is vastly larger than the number of query terms ($N \gg n$). As the set-based model dictates, the ranking scheme is expressed as the ordered cosine similarity between the collection documents and the query.

$$sim(Q, d_j) = \frac{\vec{d}_j \cdot \vec{Q}}{\|\vec{d}_j\| \times \|\vec{Q}\|} \quad (8)$$

2.4 Dense Retrieval

The current flow of ranking research in information retrieval ([Lin et al \(2022\)](#)) is on dense neural models, which use contextualized representations of terms and documents through deep machine learning techniques. The most important one is the BERT model of [Devlin et al \(2018\)](#). It is a large language model (LLM) exploiting the architecture and the attention mechanism proposed by [Vaswani et al \(2017\)](#). The focal point of BERT is the training on a vast amount of data on two separate problems, the Masked Language and the Next Sentence Prediction. Diving a bit deeper into the model, it takes as input a sequence of document terms and offers as an output a vector of embeddings for each term, as well as, a vector for the document, known as classification (CLS) token, since it is mainly used for classification purposes. The attention mechanism highlights the important parts or document terms, thus branching away from the recurrent architecture that was widely implemented in sequence models. Therefore, the model offers a well-formed dense representation of terms and documents.

Such representations can be utilized alongside classical models on tasks like reranking ([Nogueira et al \(2019\)](#)), document and query expansion ([Mallia et al \(2021\)](#)), or even on creating new dense retrieval models, for instance, ColBERT ([Khattab and Zaharia \(2020\)](#)). The latter model exploits the BERT model and creates a scoring function that achieves improved results on various information retrieval problems. The main idea is to pass each document and query pair, through two separate and independent (BERT) transformers, embedding into term vectors information from the model. For each query and document term, a cosine similarity is calculated creating a matrix of similarities for query and document terms. For each query term, the max similarity is calculated. To be more specific, for each row in the similarity matrix, the maximum value is located. Those max similarities are added to create a query-document relevance score. That method and scoring function are applied to each document in the collection and finally, by sorting documents by their score, a ranking emerges.

3 Proposed Methods

The purpose of this section is to examine the rational path graph completeness, in order to refine and model more appropriately the relationship among terms in a document. At first, the edge pruning methods proposed by [Zhou et al \(2012\)](#) were considered, but due to the fear of information loss, a windowed approach seemed linguistically more reasonable.

3.1 Windowed Rational Path Graph

In the initial formulation of the Graph-Based extension of the Set-Based model of [Kalogeropoulous et al \(2020\)](#), a graph construction algorithm was proposed having as input a set of unique terms and the respective document frequencies and calculating the edge weights using the equations in definitions 1 and 2. In our proposal, we follow an alternative approach, where the initial input is defined by the whole unprocessed

document. This algorithm is preferable because it improves the time complexity of the document pre-processing compared to [Kalogeropoulos et al \(2020\)](#).

Algorithm 1: Rational Path Graph Construction Algorithm

Input: Document D_j
Output: Graph G_j

```

1 insert each term in a Stack S
2 while  $|S| \neq 0$  do
3    $t = S.pop()$ 
4   if  $t \notin G_j$  then
5     increase the weight of the edge  $\{t, t\}$  by 1
6   else
7     initialize an edge  $\{t, t\}$  with weight 1
8   foreach  $x \in S$  do
9     if edge  $\{t, x\} \in G_j$  then
10      increase the weight of the edge  $\{t, x\}$  by 1
11    else
12      initialize an edge  $\{t, x\}$  with weight 1
13 ;
```

Let a document collection $D = \{d_1, d_2, \dots, d_n\}$, $n \in \mathbb{N}$ and suppose that each d_j consists of terms that exist in a dictionary $T = \{t_1, t_2, \dots, t_n\}$. For each document, the respective graph is constructed as described on algorithm 1. The algorithm is designed to create the graph, nodes, and edges with their weights, assuming that in the document, all terms are related to each other.

More analytically, assuming the following sequence of document terms $t_1, t_2, \dots, t_i, \dots, t_j, \dots, t_n$. We aim to calculate the weight of the $\{t_i, t_j\}$ edge, as shown on the algorithm 1.

If i equals j , then each time the term t_i is taken out from the sequence the $\{t_i, t_i\}$ edge's weight will be incremented by the number of t_i co-occurrences. Thus, assuming that tf_i is the term frequency of t_i , the first time t_i is popped the edge will have as weight tf_i and on the last, the already calculated weight plus 1, which concludes to a simple arithmetic sequence. If we formalize:

$$W_{in} = tf_i + (tf_i - 1) + (tf_i - 2) + \dots + 1 = \frac{tf_i \cdot (tf_i + 1)}{2} \quad (9)$$

If i does not equal j , then each time the terms t_i or t_j are popped the $\{t_i, t_j\}$ will have its weight incremented by 1, for each remaining t_i or t_j depending on the popped term. Due to the commutative law and the undirected edges of the graph, we can rearrange the term sequence to contain all the t_i terms followed by the t_j terms. Finally,

if tf_i is the number of occurrences of t_i and tf_j that of t_j , then,

$$w_{out} = tf_j + \dots + tf_j = tf_i \cdot tf_j \quad (10)$$

As mentioned previously the graph-based approaches are usually combined with a window approach, where a window is equivalent to the notion of a text segment or a sentence. However, as shown on algorithm 2, the in-document relationships, are altered when implementing text windows.

Algorithm 2: Windowed Rational Path Graph Construction Algorithm

Input: Document D_j , window size W
Output: Graph G_j

```

1 Split  $D_j$  into  $n$  parts  $P$ , with  $n = \frac{\|D_j\|}{W}$ 
2 for  $i$  in range  $[0, n - 1]$  do
3   insert each term of  $P_i$  in a Stack  $S$ 
4   while  $|S| \neq 0$  do
5      $t = S.pop()$ 
6     if  $t \notin G_j$  then
7       increase the weight of the edge  $\{t, t\}$  by 1
8     else
9       initialize an edge  $\{t, t\}$  with weight 1
10    foreach  $x \in S$  do
11      if edge  $\{t, x\} \in G_j$  then
12        increase the weight of the edge  $\{t, x\}$  by 1
13      else
14        initialize an edge  $\{t, x\}$  with weight 1
15    ;

```

It is of high importance to notice that the proposed algorithms do not implement overlapping windows. Moreover, on most windowed approaches, the size is a given constant W , thus as an alternative option, experiments were made by calculating the window size w as a percentage of the document size (eq. 11).

$$W = |d_j| \cdot p, \text{ where } p \text{ is the user defined percentage} \quad (11)$$

The main reason the algorithms 1 and 2 are presented is to better understand that the terms are connected as an undirected graph, and how their connection is formed. The time complexity of the algorithm (alg.1) is $\frac{1}{2} \cdot m(m + 1) = O(m^2)$, for every document, where m is the number of document terms. On the windowed approach 2, the complexity will be calculated as, $n \cdot (\frac{1}{2} \cdot W(W + 1))$, where n stands for the number of document partitions and W is the window size. The window size is less than the document length, reducing the graph construction time and creating graphs containing

cliques. Consequently, the windowed method deviates from completeness, providing graphs that capture the semantic, as well as, the syntactic notion of the document.

As an effort to attain a more accurate sentence perspective, an interchangeable window is proposed by using the dot symbol to locate sentences. That particular method aims to compare, in terms of performance, the perfect sentence window against the other windowed methods, that employ different rules on text segmentation. However, that showed no meaningful result to report.

Until this point, the sentence view is expressed as a single text window. As an effort to include in-paragraph relationships, a second window can be implemented similarly to that of a sentence but different in size. Finally, the importance of sentence-paragraph correlation can be manipulated by two parameters α, β , ultimately altering the edge weight according to the following equation:

$$W_e = \alpha \cdot W_{e_{sentence}} + \beta \cdot W_{e_{sentence}} \quad (12)$$

3.2 Importance of Bridge Nodes

Following the work of [Tixier et al \(2016\)](#), [Rousseau and Vazirgiannis \(2015\)](#) as well as that of [Kalogeropoulou et al \(2020\)](#) an effort to elaborate on the main core nodes of each graph’s core decomposition is made. At first, we try to amplify important nodes as shown in the above-mentioned related work. However, we noticed that in our proposed method, the main core consisted of not only keywords but also stopwords. That acted as a motivation for further research on the stopwords detection problem.

For each Rational Path document graph, we implemented the core decomposition scheme and noticed that on average 30% - 40% of nodes that consisted of the main core were stop-words. That of course is a strong indication, that the main core contains mainly stop-words, keywords, and a few words of random unknown importance. Implementing the same method on the union graph, the percentage of such words was 67% on average,

Taking all the above into consideration, we proposed a stop-word detection algorithm as described in algorithm 3, In particular, we employed a sample of the document collection, plus its respective graph representations, and we proposed a scoring function S_i for each graph node i :

$$S_i = \frac{\sum_j^N e_{i,j}}{W_{size}} + \alpha \times W_{p_i} \quad (13)$$

where, $e_{i,j}$ is the edge weight for every edge that is related to node i , w_{size} is the window size that normalises the edge sum and W_{p_i} is a binary sum that expresses the existence of node i in a document graph. By sorting accordingly, a ranking emerges locating the top stop-words.

3.3 Ensemble and Reranking Methods

To maximise the effectiveness of our proposed approach and exploit its benefits, we considered its combination with other models, like BM25 ([Robertson et al \(2004\)](#))

Algorithm 3: Stop - Word Detection Algorithm

Input: Collection C
Output: Word Ranking List L

- 1 Sampling (i.e hashing by Document ID)
- 2 **for** $d_i \in C$ **do**
- 3 Assign d_j to bucket: $Y = ID_{d_j} \bmod X$
- 4 **for** $d_k \in BucketY$ **do**
- 5 **for** term $t_i \in d_K$ **do**
- 6 $S_i = S_i + \frac{\sum_j^N e_{i,j}}{W_{size}} + 1 \times \alpha$

and ColBERT (Khattab and Zaharia (2020)). We employ separately the approaches of ensembling and reranking, as we describe in the text that follows.

To exploit the advantages of some implementations and cover some of their shortcomings, an ensemble technique was used to fuse model results into one final model. In continued-valued cases, the most popular ensemble option is the average or even weighted one. We employ a well-known voting method, termed Borda Count (de Borda (1781)) as described in the following algorithm:

Algorithm 4: Borda Count Algorithm

Input: Ranking A, Ranking B
Output: Final Ranking

- 1 candidate set = Ranking A $\cup$ Ranking B
- 2 **for** $candidate \in candidate\ set$ **do**
- 3 candidate scores[candidate][score] = 0
- 4 Points of each candidate will be determined by the respective index;
- 5 **for** $candidate \in candidate\ scores$ **do**
- 6 **for** $can \in Ranking\ A$ **do**
- 7 candidate scores[can][score] += |RankingA| - index of can in $Rankin\ A$
- 8 **for** $can \in Ranking\ B$ **do**
- 9 candidate scores[can][score] += |RankingB| - index of can in $Rankin\ B$
- 10 Sort by candidate score by score on a **Final Ranking**

We consider models as voters and documents as candidates. The algorithm gets as input the base model rankings and outputs the final result list. The algorithm exploits a two-voter case for simplicity but it can be generalized to accommodate more models. The candidate set is created as the union of every ranking, handling any case of different candidates in the ranking list; then the score for each candidate is initialized with a 0 value. Finally, if we consider that the most important candidate will be on the left side of the list indexed by 0, the score for each document in the ranking will be calculated as the difference between the list's size with the candidate's index. That

score will be added to the total candidate score. When all documents and rankings are processed, the final result will be presented, by sorting the candidate list by score.

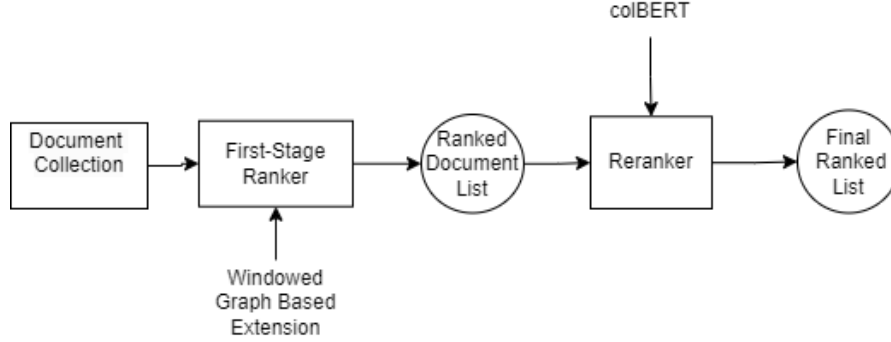


Fig. 2 Reranker Schema.

In the case of neural models, such as colBERT, another approach is usually used, that of reranking. As the most simple and effective approach, initially, a base model handles the ranking process as a first-stage ranker and thereafter the reranker, which in our case is the colBERT model, selects and rearranges the upper part of the ranking list accordingly. That process is depicted in figure 2. Hence in our evaluation, we will try to estimate the benefits of our approach, when used as a first-stage ranker.

4 Experiments

4.1 Collections and Experimental Setup

This section focuses on the experiments¹ on multiple collections ([University of Glasgow \(2011\)](#), [Shaw et al \(1991\)](#)).

Name	Acronym	Docs	Queries	Size (in MB)
Cystic Fibrosis	CFC	1.239	100	1.47
Time	TC	423	83	1.5

Table 1 Collections and types of experiments used on each one.

As a baseline, the simple set-based and the BM25 model were used. For each query in the collection, the Average Precision (A.P) is calculated. In each experimental case, the mean of average precision (MAP) is depicted, as shown in the following figures. The system that was used consists of a Ryzen 7 4800H 8 Cores (16 Logical) with clock speed at 2.9 GHz, and 16GB of RAM on a Windows operating system.

The Cystic Fibrosis ([Shaw et al \(1991\)](#)) and Time ([University of Glasgow \(2011\)](#)) collections were used for the evaluation of the retrieval ranking problem, and the Time

¹Our source code is freely available at: [Github GSB models](#) and [Github wGSB](#)

collection alone for the stopwords detection. Such collections offer queries and relevant documents judged by experts for metrics calculation. The Time collection also offers an expert-derived stopwords list allowing for validation of our method. On the other hand, Cystic Fibrosis consists of 1239 well-formed abstracts and short or at maximum medium-sized queries. Finally, the time collection offers large documents accompanied by medium to large-sized queries.

Our choice to focus on this collection choice was made by taking into consideration the variety of offered documents. Therefore, our models are evaluated on collections with different characteristics (see Table 1).

4.2 Trade-off between Processing time and Precision

In this section, we will evaluate the effect and importance of user-specified support on frequent set extraction, a technique that is borrowed from association rules algorithms. Figure 3 depicts the time needed to index the Cystic Fibrosis collection, and to answer the respective 100 queries. It is reasonable to notice that as the support gets lower, the higher becomes the processing time (in seconds); however, more information is engulfed in the document vectors, since they become larger. This implies that all models based on the set-based approach will perform better in precision on small support values, but with a price in time.

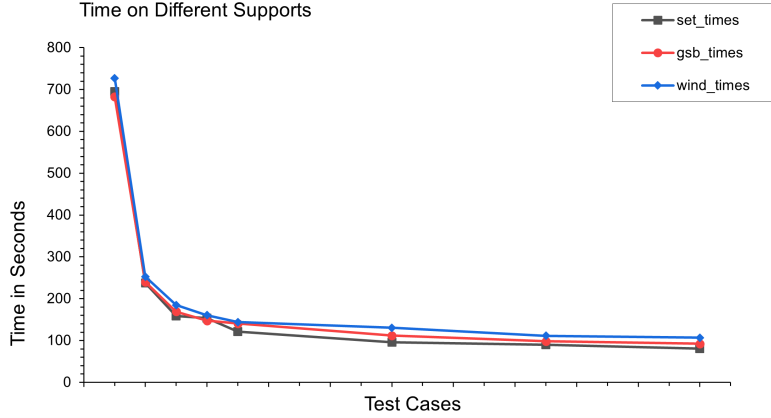


Fig. 3 Time Difference on Supports on CFC.

In figure 4, our intuition concerning the improved precision for small support values is validated. An observation can be made that the performance loss is close to linear as the support increases, with the windowed approach having the best performance throughout the graph.

However, it is important to define a support value for which the compromise between precision and processing time is optimal. We propose a simple weighted average metric as defined on equation 14. That support metric considers the total model

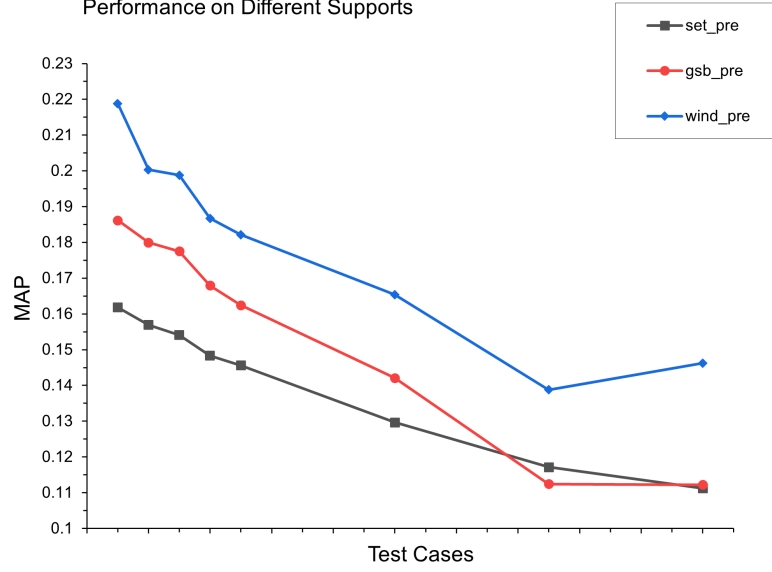


Fig. 4 Performance on Supports on CFC.

	Model	MAP
1	Set Based Model	0.161875265
2	Graphical Extension of SB	0.186186087
3	Windowed Extension	0.218826299
4	BM25	0.222656198
5	Graph of Words	0.117287827
6	Windowed Model Boosted by Keywords	0.21143052
7	Ensemble Windowed Model-colBERT	0.231127416

Table 2 Comparison of Models on Their Best Case at CFC.

time (indexing and query answering), as well as, the MAP of the model.

$$Support\ metric_i = \alpha \cdot \frac{1}{total\ time} + \beta \cdot MAP\ of\ model_i \quad (14)$$

The parameters α and β are importance variables, that sum to 1. Intuitively, we consider the precision of the model as more important, thus β equals 0.7.

Excluding the $Support = 1$, value, a proposed range for the support as depicted by the metrics graph is $[3, 5]$. However, since we wanted to perform a large in size set of experiments escaping large time delays, we exploited the similarity of the loss in performance of the set-based extensions, and we conducted the experiments with support equal to 10.

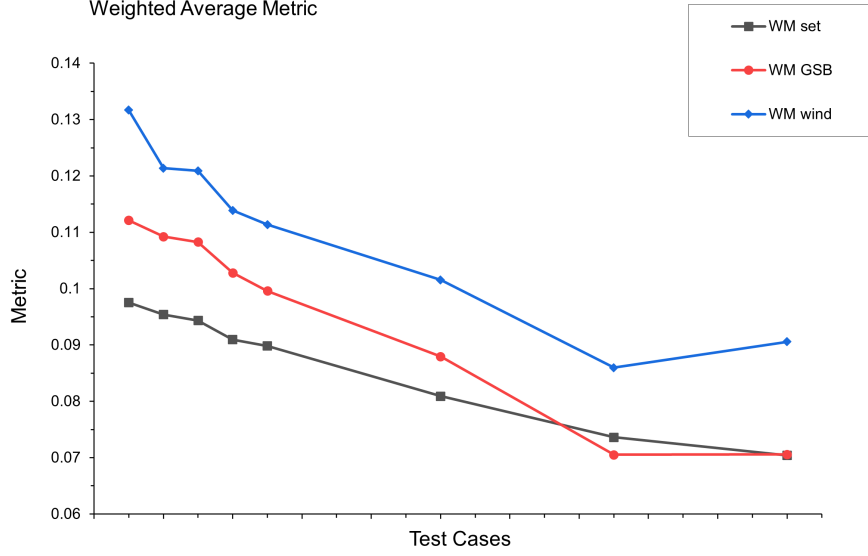


Fig. 5 Support Metric on CFC.

4.3 Models Performance on Ranking

As table 2 shows, in the best case, having fine-tuned the support and the window size the BM25 outperforms our proposed models by 0.003829899. That means that our approach is on par with one of the state-of-the-art models on the sparse retrieval aspect. The percentage windowed extension performs almost the same as the constant windowed, thus it is omitted. Emphasizing the queries, by comparison of each average precision, the BM25 model shows greater results on 56 queries versus 44 of the graph-based extension. In every experiment conducted, only the title of each document and the text (i.e abstracts) were considered as valid information. Any other provided meta-data were excluded. Therefore we can conclude that our experiments on the Cystic Fibrosis collection are according to the reported data (Rivas et al (2014), Trotman (2004), Trotman (2005)) or even better, since the document collection is slightly pre-processed. Stemming-lemmatizing and stopwords filtering operations were omitted in order to gauge the impact on the models' performance. As we will discuss in a later subsection stopwords are crucial for graph-based extensions, especially when windows are applied.

The notable difference between each model's precision is in well-formed queries with small relevant document lists. A well-formed query will be considered a query with a reasonably large amount of impactful terms. In such queries, the BM25 model shows a slight superiority. On the other hand, the graph-based extension shows a slightly improved performance on very small queries bloated with stopwords, or on very large queries with an abundance of relevant documents.

Furthermore, the graph-based extensions allow us to elaborate on other NLP tasks besides the retrieval, which is an advantage over the simple BM25. It is important

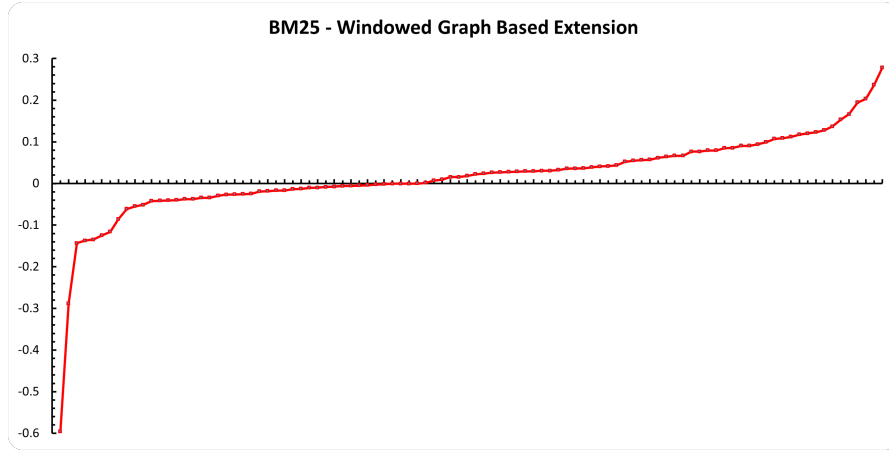


Fig. 6 Comparison of BM25 with the Graph-based Extension on CFC.

to notice that the set-based model's performance increases when amplified by the aforementioned extension.

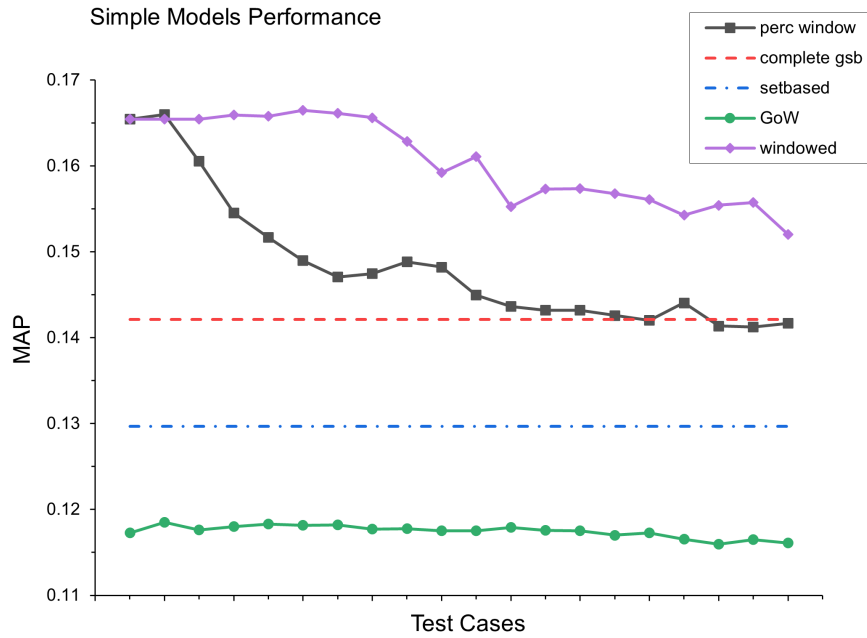


Fig. 7 Precision of Proposed Models on CFC.

In figure 7 the MAP of each model is depicted on the Cystic Fibrosis collection. As we move away from the start of the x-axis the window size is increasing. Therefore for very small values on the percentage case, the constant and percentage window

cases perform the same. On the other hand, on large values, the percentage model tends to perform as the complete graph-based extension of the set-based model (GSB) (Kalogeropoulos et al (2020)).

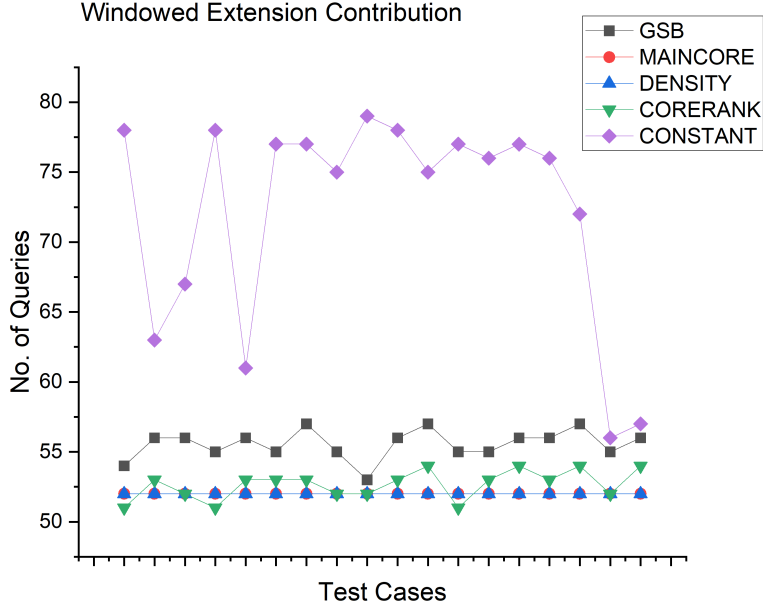


Fig. 8 Difference with the Set-Based Model of Proposed Models on CFC.

To further amplify our case, taking into consideration that the MAP metric could be misleading, we counted the number of queries that our proposed models outperformed the set-based model (Possas et al (2002)). That is expressed by the difference in average precision for each query between the set-based model and the rest, as it is depicted in figure 8. The windowed case achieves improved results at 81/100 queries. The same figure depicts the behaviour of models implementing keyword highlight proposed by Kalogeropoulos et al (2020), Tixier et al (2016), and Rousseau and Vazirgiannis (2015).

For reproducibility, the same process is adapted to the Time collection (University of Glasgow (2011)). Figure 9 depicts the difference between the proposed models and the set-based. A noticeable observation of this collection stands on the sentence-paragraph extension, which allows us to conclude that in collections with large documents that model produces positive results. In any other case, due to the small size of the texts, the model’s behaviour tends to be similar with the complete one.

Finally, in figure 10, we elaborate on the importance of windows and pruning edges on the Union Graph. We consider the complete case and then remove edges, with

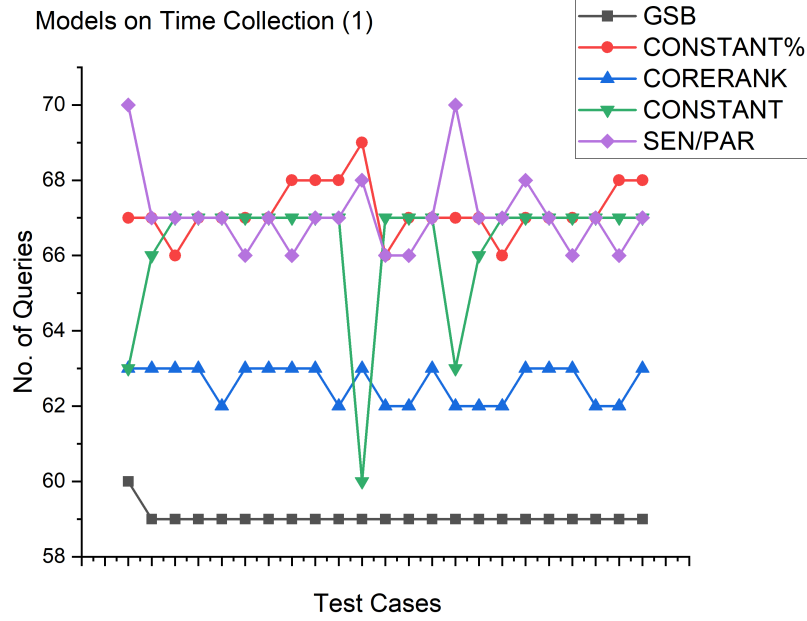


Fig. 9 Difference with the Set Based model of Proposed Models on TC.

weight, less than a threshold for each experiment on the Cystic Fibrosis collection. That process allows for more coherent graphs, whose information can be implemented in various applications, such as node-terms conceptualization via graph clustering algorithms. The community detection problem on textual graphs can be transferred as a word semantic proximity detection issue.

The model performs better in capturing the relationship among terms in the collection, when the Union Graph consists of less than 500.000 edges, thus concluding that any form of meaningful edge removal is deemed necessary on the complete graph case.

4.4 Performance of Stopword Detection Algorithm

Before the stopwords detection solution, we conducted experiments, to gauge the impact of such words in our model. We concluded that they positively affect the process. The main and most probable reason is that graphs, that contain stopwords capture the syntactic notion of the text, but also act as a hub node, that belongs to many cliques, thus not allowing the rational path graph to be separated on complete subgraphs on a large document scale.

The Time collection offers a stopwords list, which contains 340 expert-chosen words. We sampled that collection by hashing the documents' IDs into buckets of text. Thereafter, we implemented the ranking proposed by equation 13. The window size was chosen as 6 or 9, which yields the best results on the specific collection, and the sample size was 12%, 16%, 25%, and 33% of the total collection documents.

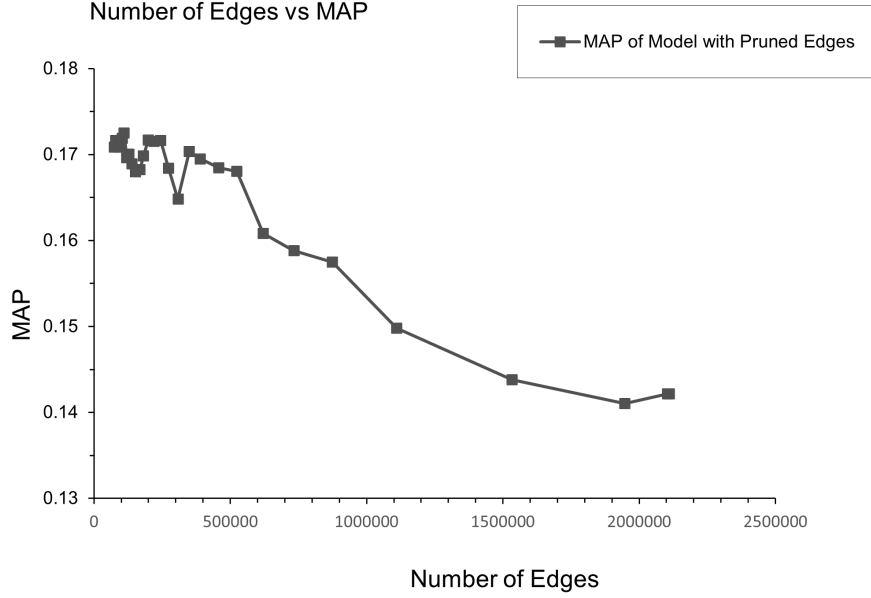


Fig. 10 Effect of Number of Edges on MAP.

No of Words	Sample 1	Sample 2	Sample 3	Sample 4
10	0,9	0,9	0,9	0,9
25	0,96	0,96	0,96	0,96
50	0,96	0,94	0,94	0,96
100	0,9	0,89	0,87	0,87
200	0,72	0,73	0,71	0,725
340	0,61	0,63	0,57	0,59
450	0,52	0,54	0,50	0,508
1000	0,287	0,287	0,289	0,282

Table 3 Precision for Collection Samples on TC

Table 3 depicts the precision of stopwords at the k top words of our ranking. In that particular case, the number of samples was 6, the window size was 9 and the parameter α was 5. It is important to notice that at the first 100 words, our method dictates 90% of the collection stopwords. Moreover, the behaviour among the samples is similar. Therefore, we conclude that one sample can represent the collection on the stopwords detection problem. The difference, or the lack of it, among the samples is definite enough in Fig. 11.

4.5 Performance of Ensemble and Reranking Methods

On the information retrieval model combinations, we made an effort to deviate from the reranking aspect of the current research tendency to model ensembles; hence we experimented with both alternatives. In chapter 3.3, a simple algorithm based on

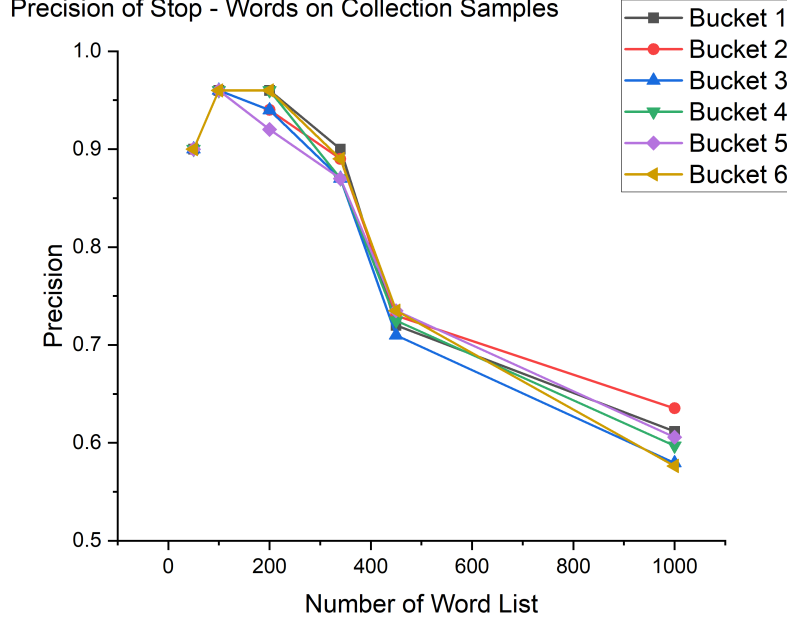


Fig. 11 Stopword Detection Precision on Collection Samples.

Borda count was presented, which will be implemented in the following experiments, for ensembling.

The focal point of our attempt is to estimate empirically, the impact of fusion models implementing graph textual representations, as well as the effect of such models in approaches that are considered state-of-the-art such as the colBERT ([Khattab and Zaharia \(2020\)](#)). To be more specific, our goal is not to propose an ensemble that will outperform the state-of-the-art, but a method that will restrain the model when it fails to perform in some queries.

Staring at the first issue at hand, the performance of the ensemble is depicted in figure 12. Every model that implements association rules mining for termset creation has support defined as 10. the simple unique models are considered as the baseline for comparison. An important observation is that the ensemble of the set-based model with the constant window has a smaller decline slope. That observation, though, is more important in the ensemble of models with different constant window sizes. That ensemble could be considered window agnostic, achieving the highest overall performance on larger window sizes, although, on small window size the simple window case is superior. Such observation allows our proposed models to be implemented on collections without the need for parameter fine-tuning.

As mentioned before, the window size parameter is regarded as of crucial importance, and its fine-tuning can render a model obsolete, if the parameterization is hard. Therefore, the size defined by a percentage of a document size or a multiple-windowed ensemble approach can be considered as a solution to that issue.

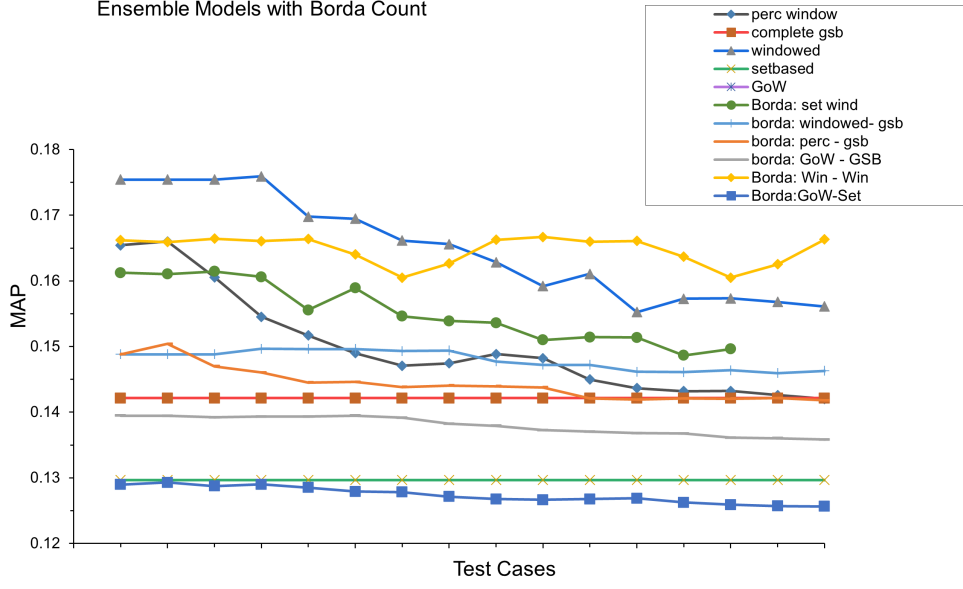


Fig. 12 Precision of Ensemble Models on CFC.

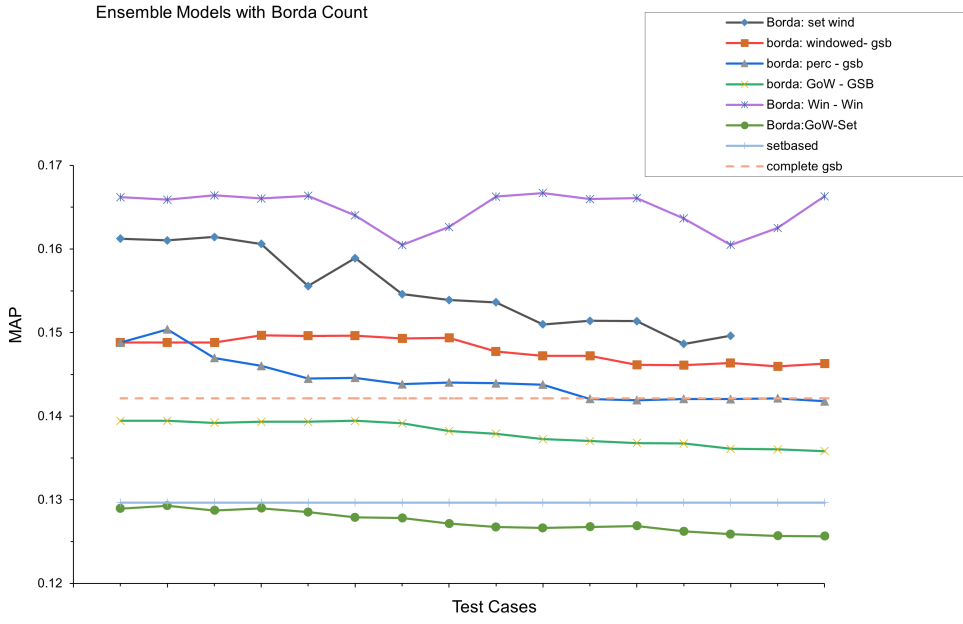


Fig. 13 Precision of Windowed Ensembles on CFC.

figure 13) elaborates on the performance of the fusion models. We clearly can validate the aforementioned observations about the newly found stability of the windowed

models. However, the rest of the fusion models tend to have similar performance to the inferior model.

In this paragraph, ColBERT’s performance will be discussed in comparison with the BM25 model and our approach. We calculated the average precision difference between colBERT and BM25 and our windowed approach, with window size of 7, on the Cystic Fibrosis collection. The BM25 model overall outperforms the colBERT at 60 queries and our approach at 50 queries. As figure 14 depicts, the graph-based extension on the superior cases against the colBERT is more effective than the BM25. Although the reverse statement has its merits. To simplify the above, when the BM25 is superior to the colBERT, its performance is better than the windowed approach (wGSB), and when the windowed model is more effective than the colBERT is more effective than the BM25 model as well.

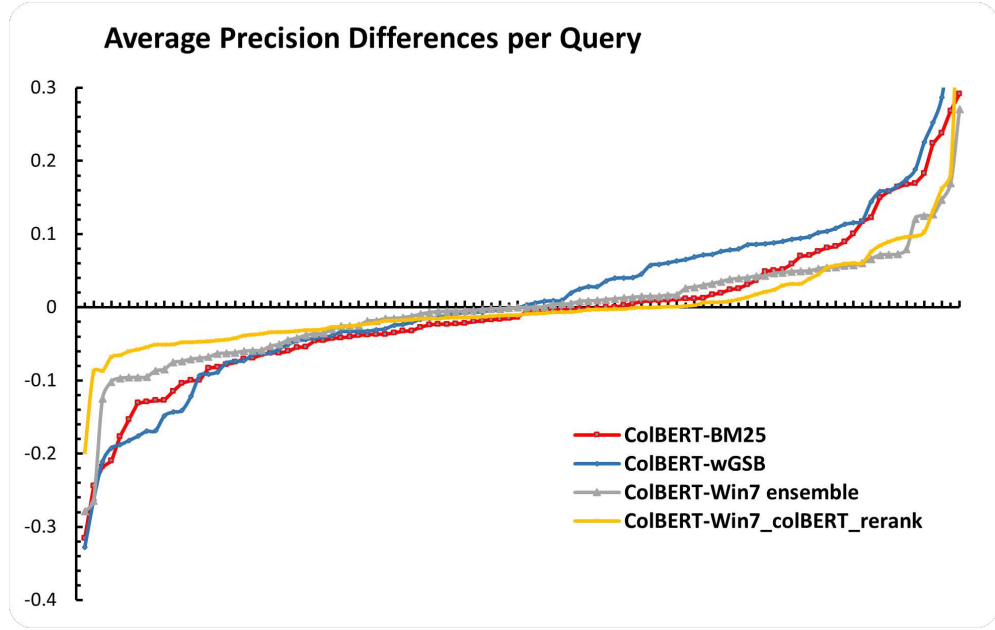


Fig. 14 Comparison of BM25 and Windowed Graph Based Extension with ColBERT on CFC per Query.

On the reranking aspect of the experiment, we implemented as a fist-stage ranker the windowed graph-based extension of the set-based model, having window size at 7 terms defined. We consider that the first 400 documents in the sorted list are the most relevant for each query. Therefore, a new collection containing those documents was the input for the colBERT model, handling the reranking process.

Figures 14 and 15 depict that the rerank and ensemble methods capture the positive impact of the ColBERT model, thus producing better results than the windowed approach in cases where the ColBERT outperforms the rest of the models. Therefore our proposed windowed method is capable of improving performance, especially when

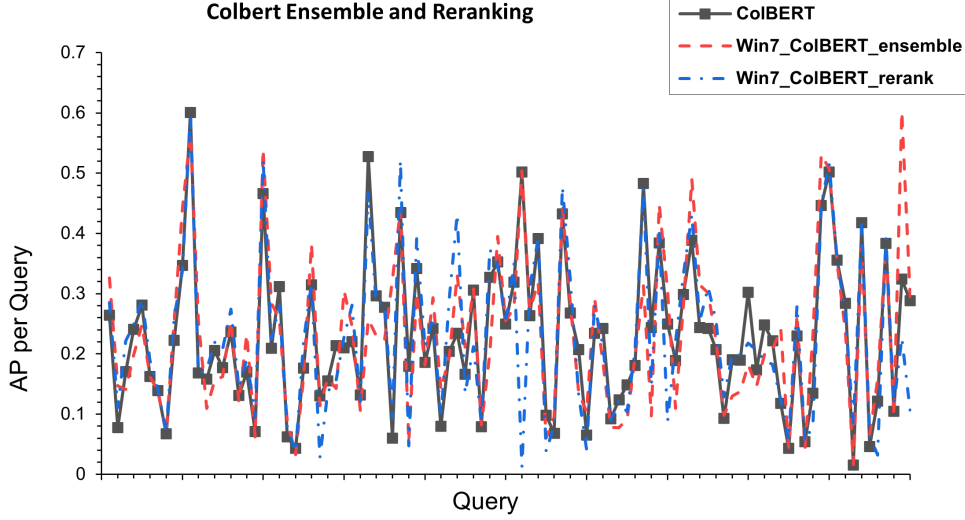


Fig. 15 Comparison of ColBERT on Ensemble and Rerank Tasks on CFC per Query.

used in ensemble and reranking purposes, providing information crucial to the retrieval process.

To validate further our approach, we tested the statistical significance of our results by conducting 2-tailed paired t-tests; our tests produced small p-values, in the worst case at most 0.000952339.

5 Conclusions and Future Work

In this paper, we proposed improved methods for a graphical extension of the set-based model, elaborating on the importance of textual segments, in such algorithms. We focused on a variety of related options, such as simple windows, a single window, or multiple, that capture the sentence-paragraph relation. Moreover, by considering the start and end of a window as the punctuation point we indicate that a strict sentence approximation might be futile. Finally, we approached the terms or node importance on a graphical aspect, classifying irrelevant terms as stopwords, on collection size, having a 90% precision on the first 100 purposed stopwords.

Our model is in parallel with a BM25, a well-known and widely used model for Information Retrieval. To be more specific, it is implemented as a first-stage ranker on models that apply transformer-based reranking in later stages. Therefore, our approach, as BM25, can be used as such. We evaluated the performance of our approaches in multiple collections with a variety of characteristics. An attempt is made to elaborate on the effect of each model's parameter and proposed methods and thought processes for choosing suitable values. Finally, a framework of model ensemble is proposed to boost the stability and effectiveness of our approaches.

In further research, the focal points should aim at decreasing the complexity of the proposed algorithms by modifying them or implementing parallelism through a distributed system, that will allow for more graphical processing. Furthermore, the node embeddings proposed by Grover and Leskovec (2016) or even word representations found in the work of Mikolov et al (2013) allow information encoding on real-valued vectors. By inducing such on-edge weights, structural and semantic information will be taken into consideration by our model. Moreover, embeddings will allow machine learning models to be applied simultaneously with graph features, such as centrality measures or core/truss numbers. Finally, the research focus has turned more to deep learning methods and algorithms, thus an effort to employ Graph Neural Networks (GNNs) (Wu et al (2022)) on graphs derived from textual data can be made, in order to exploit nodes' structural and semantic relationships.

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